Multi-Channel Agile Comb Generator for Antenna Radiation Pattern Measurements

K.E. Kolodziej and P.T. Hurst
MIT Lincoln Laboratory
Lexington, Massachusetts, USA

***Abstract*—Antenna radiation patterns are typically measured using network analyzers, which are both expensive and physically large in size. These drawbacks can limit this test equipment's usage in universities that cannot afford to make such a purchase, as well as within applications that require mobile antenna measurements. An alternative approach is to combine a low-cost receiver with a flexible signal source. This paper presents the design of a tunable comb generator prototype that is capable of outputting frequencies up to 4 GHz. The compact nature of this source along with its potential to be dynamically reconfigured yields a device that can be used to measure antenna patterns for many different applications.**

I. Introduction

The process of taking radiation pattern measurements of wideband antennas can be expedited with the use of signal sources that either sweep frequency very quickly or output multiple frequencies simultaneously. For applications that require superior precision, the pattern must be sampled with many closely-spaced frequency points or interpolated in post-processing [1]. Many of these measurements are currently taken using a network analyzer, which rapidly sweeps a continuous wave (CW) signal over a defined frequency range. While such instruments can provide fine frequency spacing over a wide frequency range, they are typically very expensive and physically large in size. These drawbacks limit their practicality in the areas of university research and mobile testing environments.

One approach to overcome these challenges is to use an agile comb generator, which can not only output multiple frequencies at the same time, but also quickly tune to different bands of interest. Such a device could be designed to have a fine frequency spacing over a narrow bandwidth and dynamically change its center frequency. An even finer grid could be synthesized if the change is a fraction of the tone spacing, or entirely different bands could be measured with larger frequency adjustments. In either case, this signal source could be inexpensively fabricated and physically compact, which increases its usefulness. In order to complete the measurement system, this generator must be paired with a receiver, which can range from a spectrum analyzer to a low-cost software-defined radio [2], [3].

This work was sponsored by the Department of the Army under Air Force contract FA8721-05-C-0002. Opinions, interpretations, conclusions, and recommendations are not necessarily endorsed by the United States Government.

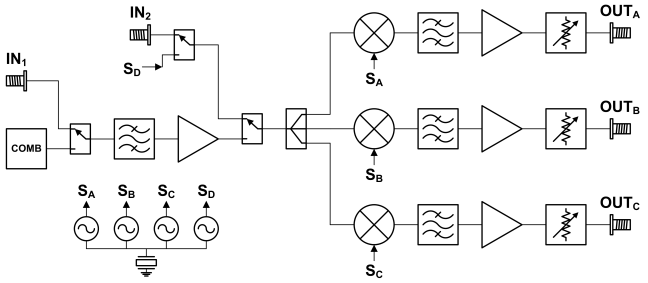


Fig. 1. Diagram of the multi-channel agile comb generator.

Section II of this paper describes the design of an agile comb generator device with multiple output channels that allow for additional measurement flexibility. The measured results are presented in Section III, and conclusions are derived in Section IV.

II. Generator Description

The block diagram of this agile comb generator design is shown in Fig. 1. The core signal source begins with a compact comb generator circuit, which produces frequency-domain tones that are spaced according to the frequency of an on-board square-wave oscillator [4]. A piece of its frequency spectrum is then filtered to limit the number of tones before the following amplification stages. In this prototype, the comb spacing was chosen to be 2.5 MHz, and a surface acoustic wave (SAW) filter with a 20 MHz bandwidth centered at 70 MHz was used to select the 9 frequency tones between 60 and 80 MHz. Next, the signal is equally divided among three identical channels and individually up-converted within the range of 100 to 4000 MHz by configuring Analog Devices' ADF4350 wideband synthesizers ($S_{A\text{-}C}$). Finally, these three signals are filtered to reduce spurious and harmonics, and variable attenuators are used to adjust the output power levels. In addition to this main operating mode, external signals could be applied to inputs IN_1 and IN_2, which allows for arbitrary waveform outputs. Further flexibility is provided by selecting synthesizer S_D as a CW input.

The three independently-controlled output channels enable several capabilities beyond those of a single signal source. The outputs could be combined to construct an even finer frequency grid, or extend the output instantaneous bandwidth

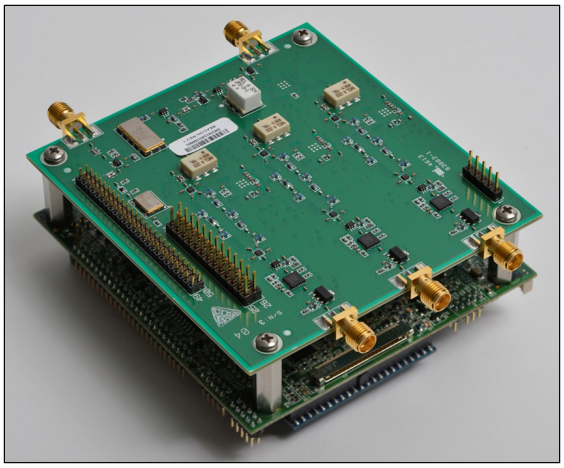


Fig. 2. Comb generator prototype board, measuring 9.0 x 9.6 cm, mounted on top of a Helios single-board computer.

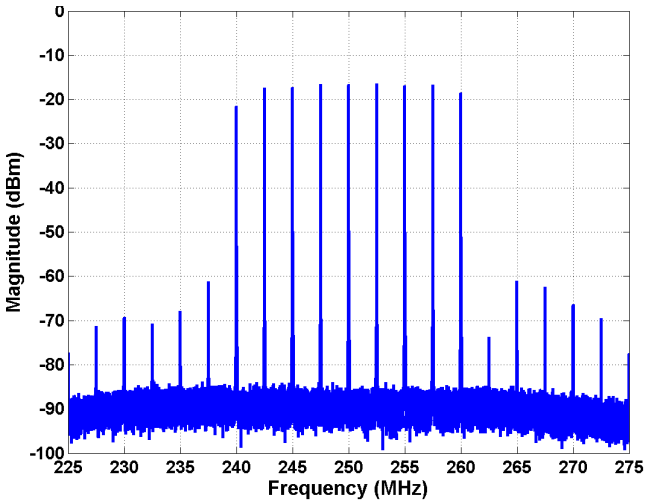


Fig. 3. Comb generator output spectrum showing 2.5 MHz-spaced tones, filtered and up-converted to 250 MHz.

beyond 20 MHz by tuning to the adjacent channels. Also, radiation pattern measurements could be made simultaneously on different polarizations if the output antennas were configured appropriately.

All of the tuning, switching and gain control was provided by a Helios single-board computer (SBC) from Diamond Systems. This board contains a 1 GHz Vortex86DX processor, 40 digital control lines, a multiplexed 16-channel analog-to-digital converter, and 4 digital-to-analog converters in a compact PC/104 form-factor. Alternatively, a Raspberry Pi could have been substituted for this SBC, but it was deemed to be a less mature and less rugged design. A photo of the comb generator prototype board mounted to the Helios SBC is shown in Fig. 2. When joined together, the two boards have dimensions of 9.0 x 9.6 x 2.5 cm, which is much smaller than a typical network analyzer. The SBC can also be used to synchronize and control the various functions of the receiver that would complete the measurement system. This control could not only be accomplished using one of the on-board serial buses or Ethernet interface, but could also be done wirelessly with the addition of a small transceiver, which would enhance the setup's flexibility.

III. Results

While this agile signal source can be configured in the many different ways mentioned above, Fig. 3 illustrates its fundamental performance, which is obtained by analyzing the single-channel output of the internal comb generator. This figure plots the output spectrum with the internal attenuator set to 15 dB, and shows the 9-tone comb, with 2.5 MHz spacing, up-converted and centered at 250 MHz. The tones have a signal-to-noise ratio greater than 60 dB and the adjacent tones are rejected approximately 40 dB relative to the desired tones, which is sufficient for most pattern measurements. The generator's total output power for the 0, 15 and 30 dB attenuation states is 6.9, -7.8 and -22.7 dBm, respectively. If additional power is required for the measurement, an external power amplifier could be added. Finally, the DC power of the generator, combined with the Helios SBC, was measured to be 8.8 W, which is low enough to support antenna tests outside a laboratory environment.

IV. Conclusion

Wideband antenna radiation patterns can be quickly measured using inexpensive signal sources that output multiple frequencies at the same time. This paper described such a source that was designed as a multi-channel tunable comb generator and was capable of providing finely-spaced tones in the frequency domain. While one operation mode of this specific prototype measured 6.9 dBm of comb output power at 250 MHz with a 2.5 MHz grid, the generator could dynamically be reconfigured to cover frequencies up to 4 GHz. Additionally, the compact form-factor and multiple output channels allow this design to be tailored to many different measurement scenarios, which include university research and mobile testing.

Acknowledgment

The authors would like to express their gratitude to Emil Eastburn and Siddhartha Krishnamurthy for technical insights, as well as David Holtman for help with the PCB design.

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